





NO MAGNETISM
WITHOUT QUANTUM.

TYPES OF MAGNETISM.

$$\vec{M} = \frac{\chi}{\mu_0} \vec{B}$$

χ = SUSCEPTIBILITY

μ_0 = PERMEABILITY

m^* NOT m

$V(x)$ DUE TO BAND STRUCTURE

// DEPLETION WIDTH

SEMICONDUCTOR
DEVICES.

QUANTUM WELL

ATOMIC MAGNETISM.

(WILL IGNORE NUCLEI)

$$\mu_B = \frac{e\hbar}{2m}$$

$$\mathcal{H} = \frac{(\vec{p} - e\vec{A})^2}{2m} + V(x) + g\mu_B \vec{S} \cdot \vec{B}$$

$$\vec{\sigma} \cdot \vec{A} = \vec{B}$$